Final Project Summary: X Education Lead Scoring Model

1. Project Overview

The objective of this project was to build a predictive lead scoring model for X Education. By analyzing lead activity, we aimed to distinguish between "hot leads" (high conversion probability) and "cold leads," enabling the sales team to prioritize outreach, increase conversion rates, and optimize resource allocation.

2. Process and Methodology

The project followed a rigorous end-to-end Data Science pipeline:

- **Data Cleaning:** We addressed missing values by replacing 'Select' placeholders with `NaN` and removing columns with >40% missing data. Rows with negligible missingness were dropped to maintain data integrity.
- **Feature Engineering:** Categorical variables were converted to dummy variables, and numerical features (`TotalVisits`, `Time Spent`, `Page Views`) were standardized using `StandardScaler`.
- **Feature Selection:** We employed **RFE (Recursive Feature Elimination)** and **VIF (Variance Inflation Factor)** analysis to eliminate noise and multicollinearity, ensuring only high-impact features remained.
- **Model Building:** A Logistic Regression model was developed, optimized, and evaluated. The model showed exceptional stability, with **93.5% accuracy** on training data and **92.8%** on test data.

3. Key Learnings and Insights

- **Performance:** The model achieved a high **Sensitivity (Recall) of 90%**, ensuring the vast majority of potential leads are captured. An **AUC-ROC score of 0.97** confirms the model's high precision in distinguishing between converting and non-converting leads.
- **Top Conversion Drivers:** Three specific variables contribute most to conversion:
    1. **Tags_Closed by Horizon:** The strongest indicator of high intent.
    2. **Tags Will revert after reading the email:** Signifies serious evaluation.
    3. **Lead Source_Welingak Website:** Consistently the highest-quality acquisition channel.

4. Strategic Recommendations

The sales team can dynamically adjust the probability threshold based on business cycles:

Business Phase	**Strategy Focus**	**Probability Threshold**	**Primary Metric**
Aggressive Phase	Maximum Coverage	**Low (~0.3)**	Sensitivity (Recall)
Target-Met Phase	Maximum Precision	**High (~0.8)**	Precision

- **Aggressive Phase:** By lowering the threshold to 0.3, we expand the lead pool. Assigning these to the 10 interns ensures aggressive pursuit of every potential opportunity.
- **Target-Met Phase:** By raising the threshold to 0.8, we filter out "noise." The sales team minimizes useless calls, focusing only on "sure-shot" conversions, allowing time for secondary tasks.

5. Conclusion

This project successfully transitioned from raw, inconsistent data to a high-precision, business-ready predictive tool. By implementing these data-driven strategies, X Education can reduce wasted sales effort, improve conversion rates, and adapt its outreach intensity to meet real-time business targets.